

Exercises in Marine Ecological Genetics

10. Metabarcoding / eDNA I

- Amplicon Sequence Variants (ASVs)
- Taxonomic assignment
- ASV abundance distributions

Martin Helmkamp

<https://github.com/mhelmkampf/meg26>



Today's objectives

By the end of this session, you will be able to:

- Inspect **Amplicon Sequence Variant (ASV) count tables**, the backbone of eDNA data
- **Classify ASVs taxonomically** using BLAST against a public reference database
- **Construct a phyloseq object** by combining ASV counts, taxonomic assignments, and sample metadata, then filter low-abundance samples and ASVs
- Visualize sequencing depth and **ASV abundance distributions**

Environmental DNA (eDNA)

- Genetic material **shed by organisms** into their surroundings
- Can be recovered from **water, soil, sediment** etc.
- Analyzed using molecular methods (e.g. PCR, DNA sequencing, metabarcoding)
- Non-invasive and **highly efficient**, can detect cryptic species
- Typically degrades over hours to weeks, usually **reflects local presence**

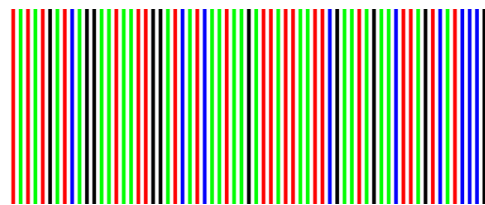
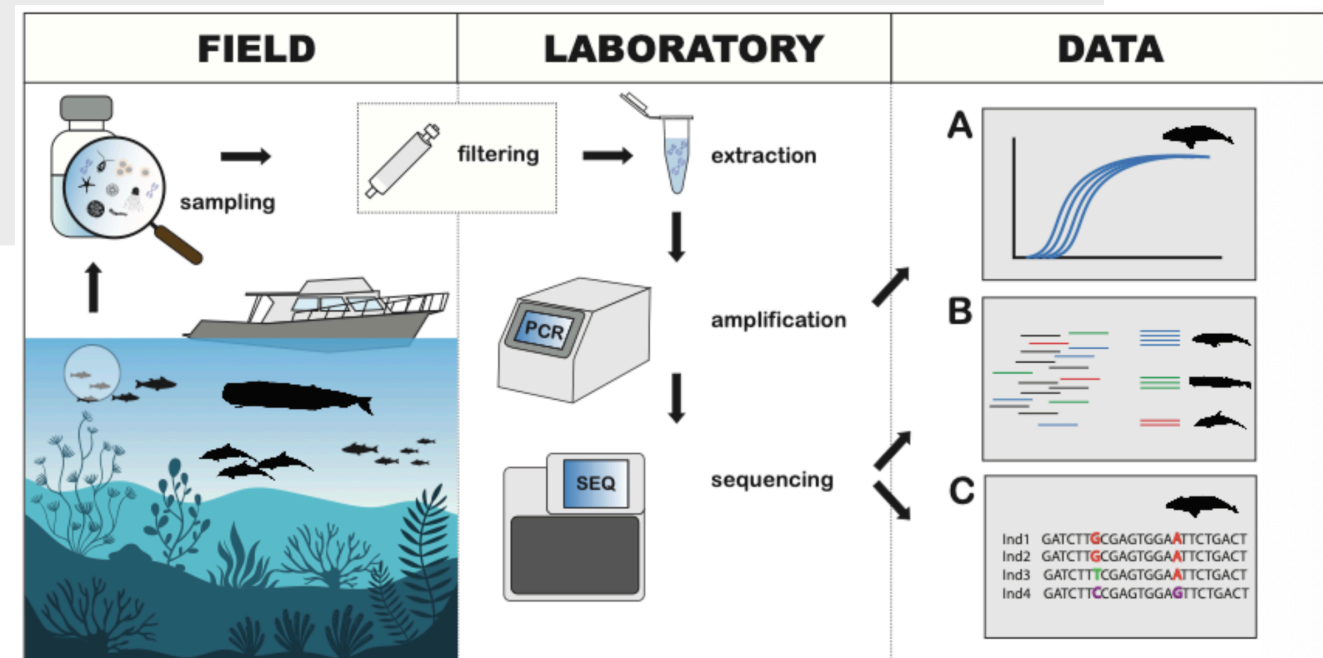


Figure by Székely et al., CC-BY 4.0

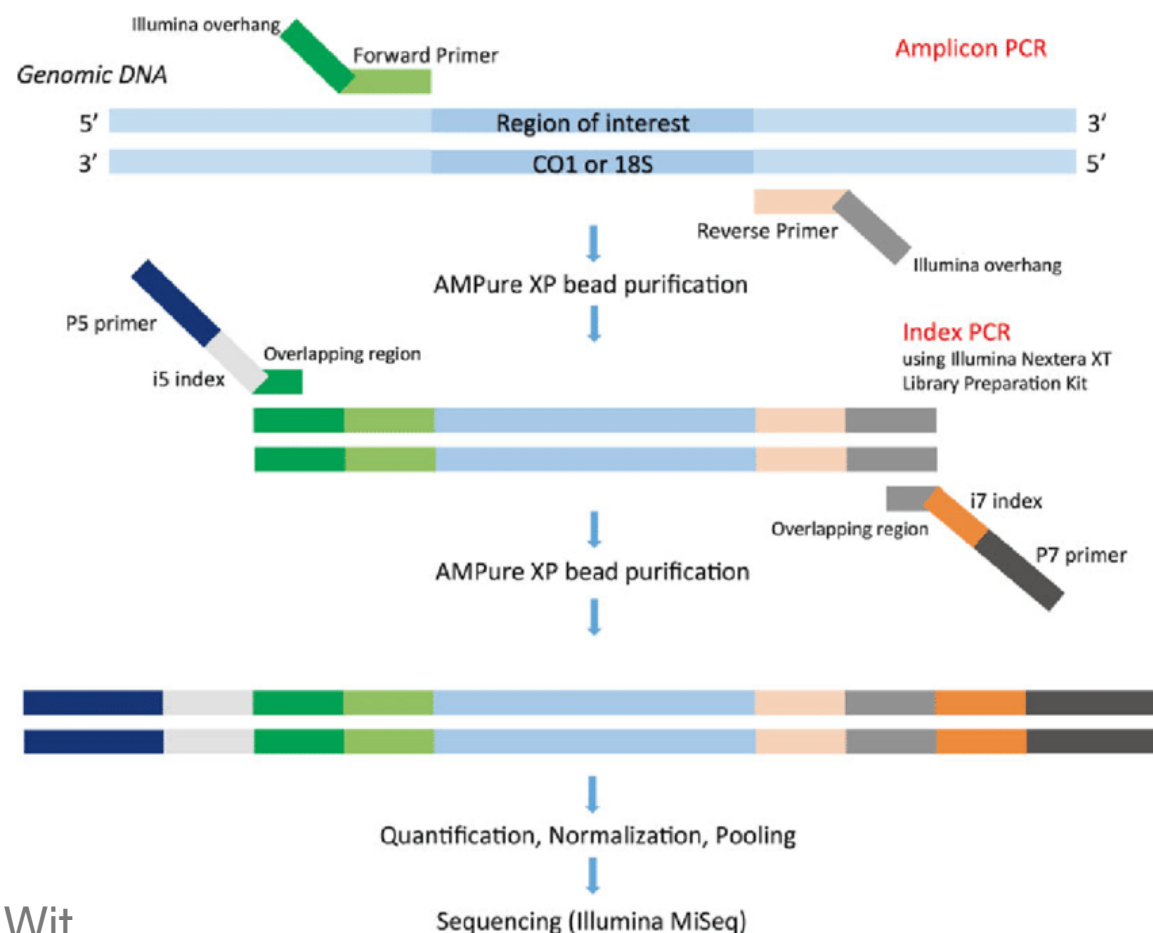


Amplicon sequencing

Next-gen sequencing of specific genomic region (e.g. 12S)

Many samples in parallel by **multiplexing** (using unique molecular index for each sample)

Dual-PCR library preparation



Pierre De Wit

Illumina MiSeq bench top sequencer

7–25 million reads, up to 2 × 300 bp



Illumina Inc.

Monitoring seagrass restoration

Dataset

- Restoration began **July 2025**
- Transplantation of single shoots and **plugs** tested at different densities
- eDNA sampling after **0, 3, 6, 12 and 18 months**



Kathy Young



Johan Hollander

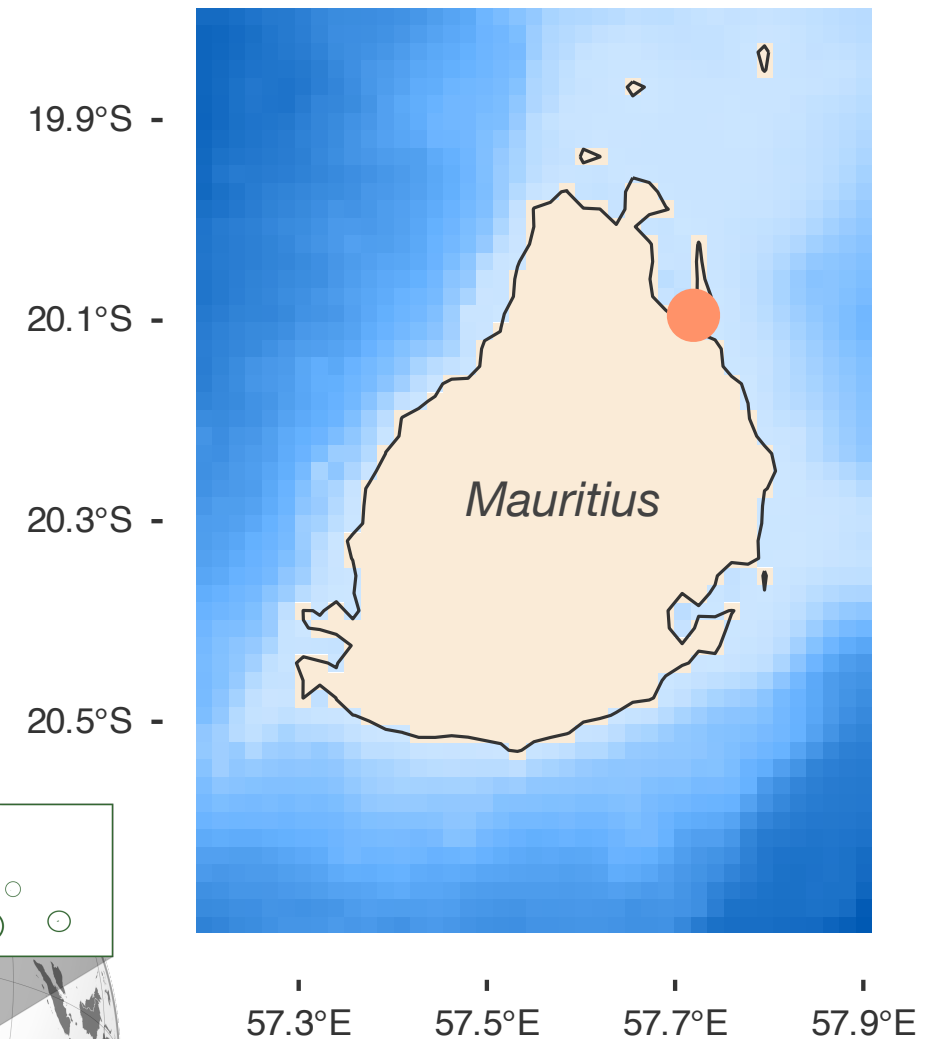
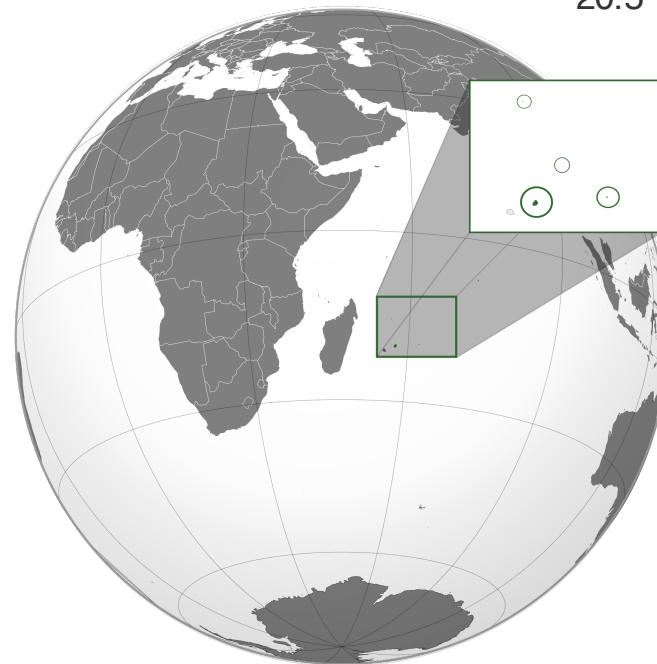


Image by Reef Conservation
Globe by Yashveer Poonit, CC BY-SA 3.0

Monitoring seagrass restoration

- Restoration began **July 2025**
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Dataset

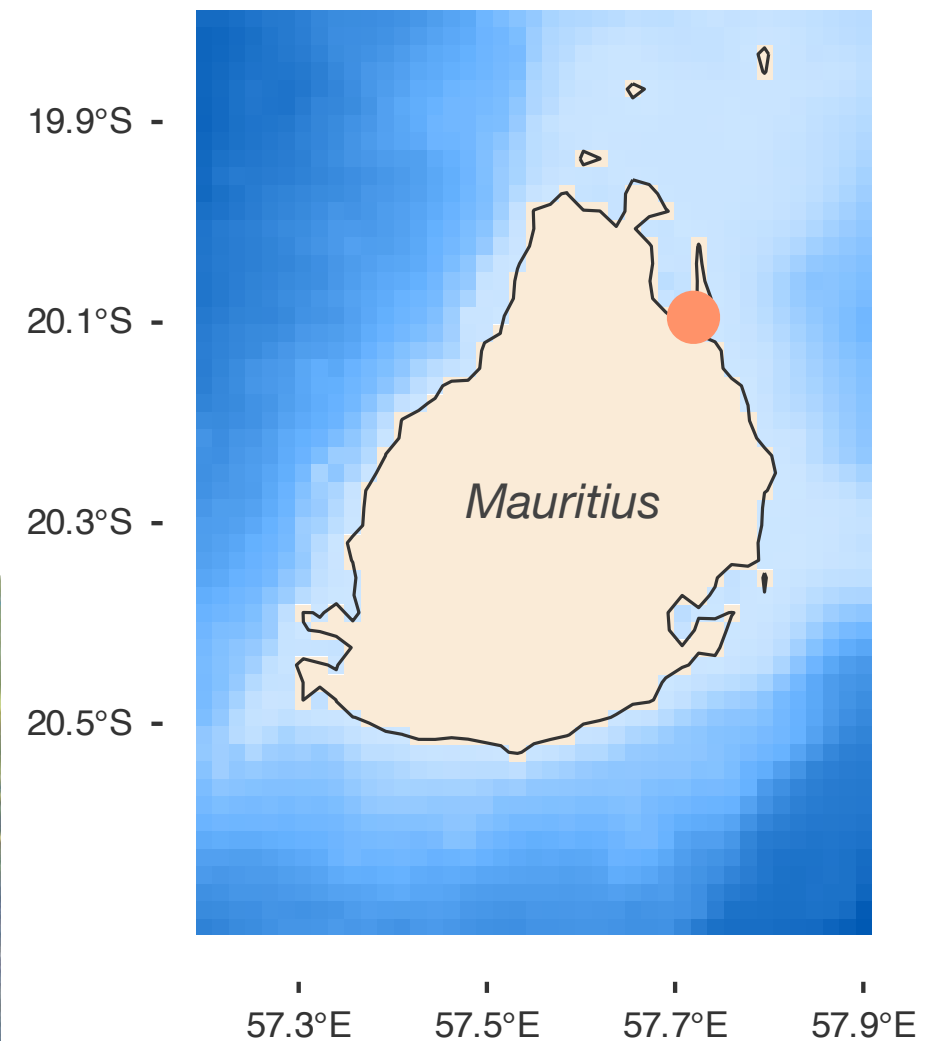


Image by Reef Conservation
Globe by Yashveer Poonit, CC BY-SA 3.0

Sampling scheme and sequencing

2 Sites

- Roches Noires
- Bel Ombre

3 Habitats

- Bare sand
- Seagrass meadow
- Channel to open ocean

2 Seasons

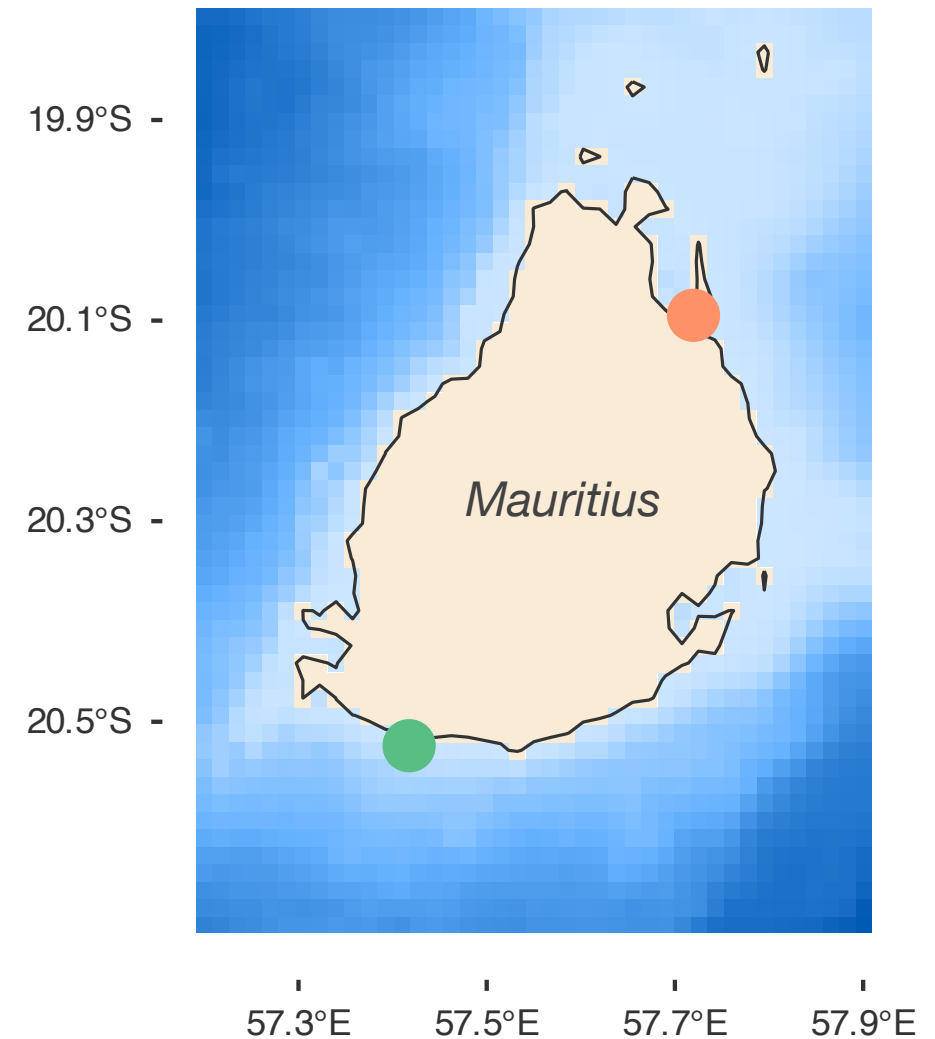
- summer
- winter

3 Replicates each

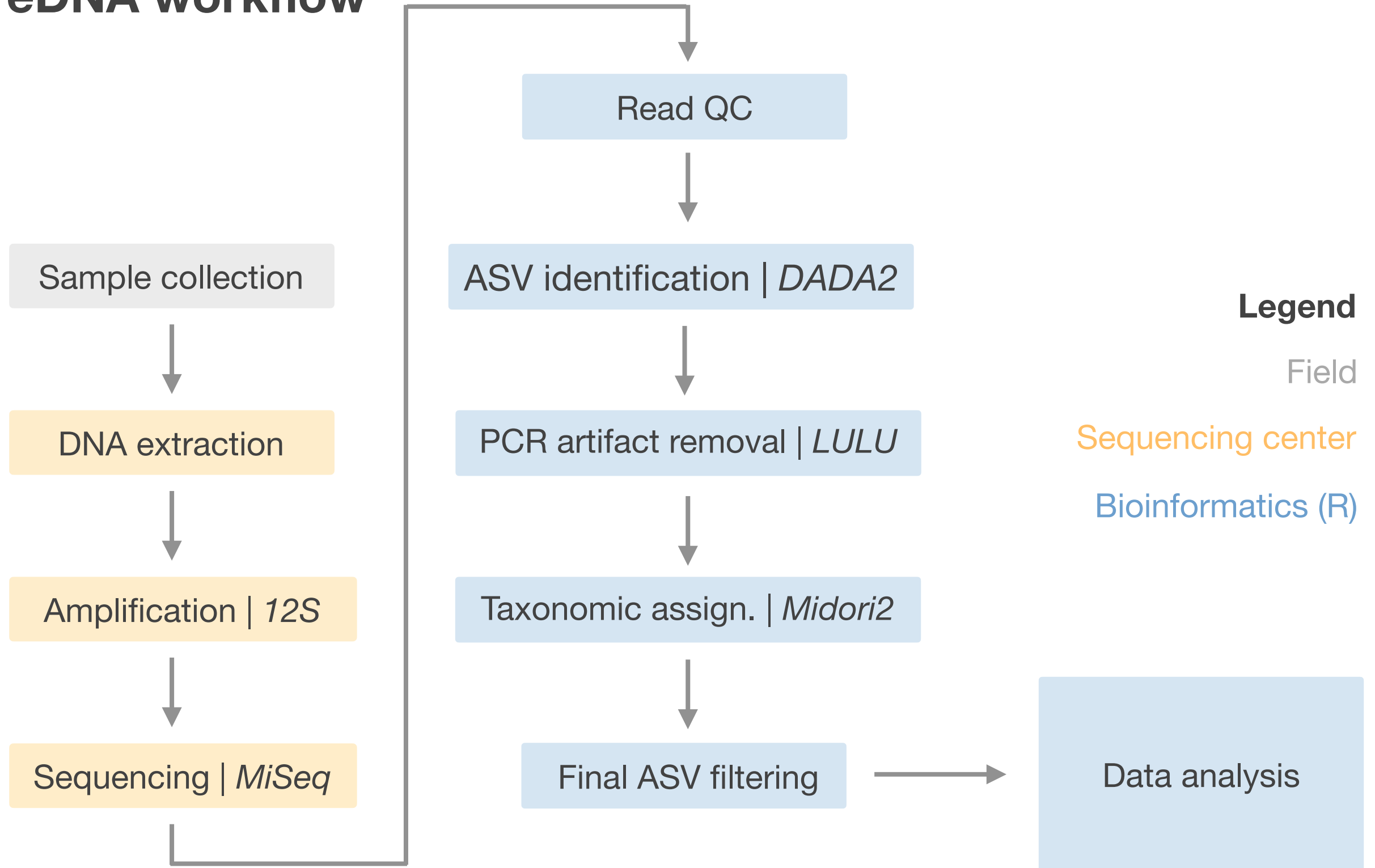
Method

- Filtered sea water
- 12S (MiFish)
- Illumina MiSeq

Dataset



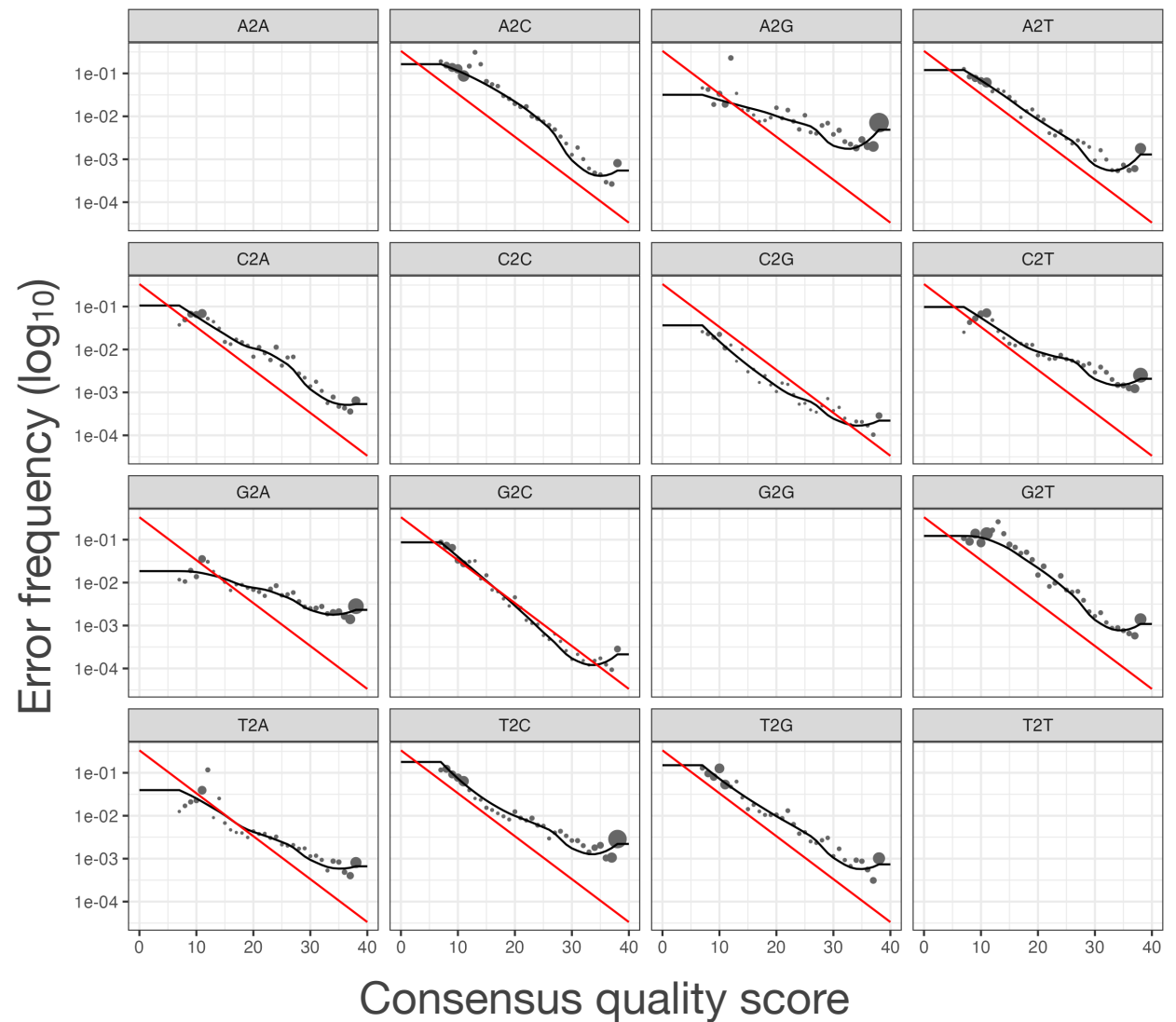
eDNA workflow



Amplicon Sequence Variant approach

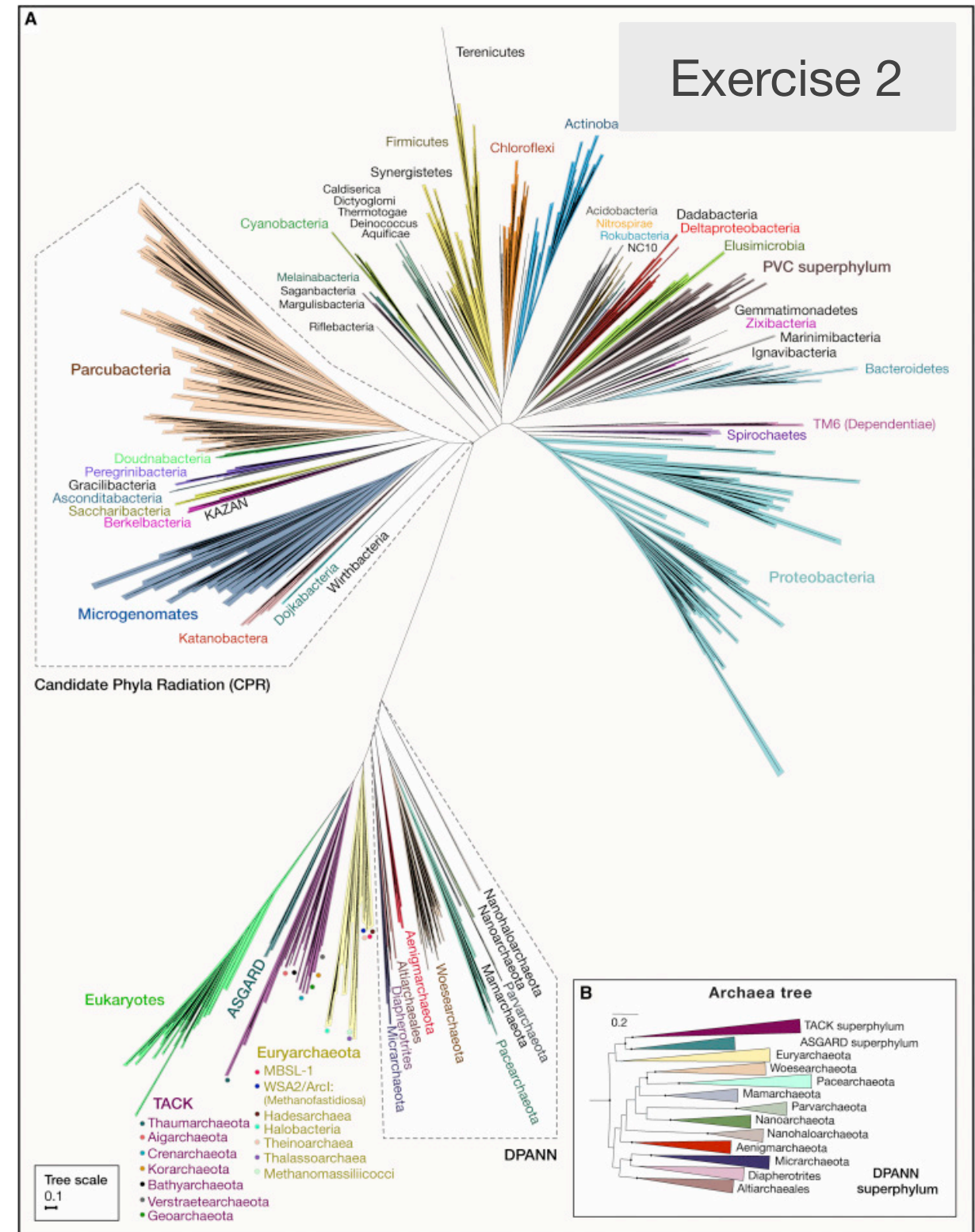
- Error rates are estimated directly from data (black dots)
- Sequences are statistically tested against the error model (black lines)
- Based on probability, sequences are discarded as sequencing errors or retained as true variants (ASVs)
- Better resolution and reproducibility than OTU clustering

Learned error rate model | DADA2



Taxonomic assignment

- Similarity / alignment-based (e.g BLAST)
- Phylogenetic placement
- Probabilistic classifiers (e.g. QIIME2)



Castelle & Banfield 2018, Cell

Matching sequences to database with BLAST

Exercise 2

Algorithm overview

- Split query into very short segments (*k*-mers or words)
- Find exact matches between words and sequences in database (seeds)
- Extend matches to **local alignments** (HSP; stop once too many mismatches occur)
- Evaluate statistical significance of each HSP (**e-value**)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
✓	Oncorhynchus keta mitochondrial COX1 gene for cytochrome c oxidase subunit 1, partial cds, isolate: OK_M08F	Oncorhynchus keta	1029	1029	100%	0.0	99.64%	772	LC094471.1
✓	Oncorhynchus keta mitochondrial COX1 gene for cytochrome c oxidase subunit 1, partial cds, isolate: OK_M01F	Oncorhynchus keta	1029	1029	100%	0.0	99.64%	772	LC094464.1
✓	Oncorhynchus keta isolate 10_Narva cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Oncorhynchus keta	1029	1029	100%	0.0	99.64%	655	KR778851.1

...

- **Clusters of similar barcodes**
- Clustering is based on genetic similarity, independent of formal taxonomy
- Represent operational taxonomic units (OTUs) that often correspond to species
- Enable species identification and biodiversity assessments

Metabarcoding / eDNA I

Summary

[illegible]

Key takeaways

Summary

- Metabarcoding uses marker genes (e.g. 12S, COI) to identify and compare organisms in complex communities
- Reads can be “denoised” into high resolution **Amplicon Sequence Variants** or more simply clustered into OTUs
- ASV count tables, taxonomic assignments, and sample metadata can be combined into a single structured object (e.g. **phyloseq**) for downstream analysis
- **Filtering low-abundance ASVs and low-read samples** is a necessary step before ecological interpretation, since raw ASV tables contain noise and uneven sequencing depth

Please help to [evaluate this course](#)

June 26 – July 17

